HumaStar 100
HumaStar 200
HumaStar 300SR
| LIS Interface Manual

Version 14



Revision list

Revision	Date	Description	Editor
1	2012/07/01	First revision	Mathias Kamprath
2	2013/08/30	References to HI software added	Mathias Kamprath
3	2016/09/21	Review of folder structure. Correction of file encoding. Example for results transmission added	Mathias Kamprath
4	2016/12/12	Correction of input file P record definition	Mathias Kamprath
5	2019/03/11	Test names note in Process description	Mathias Kamprath
6	2019/06/04	Supplement to LIS ASTM work list folders	Mathias Kamprath
7	2019/07/04	QC request input and result output added. (For HumaStar 300SR only)	Mathias Kamprath
8	2020/05/11	Adding list of not usable characters	Tony Petzold
9	2021/07/13	Added ASTM file input style	Tony Petzold
10	2022/04/07	Added detailed sequence of operation	Tony Petzold
11	2023/05/09	Minor adjustments in chapter 6 and 7	Tony Petzold
12	2023/10/24	Adjusted description of the field birthday. Added Disclaimer	Tony Petzold
13	2025/05/09	Disclaimer adjusted	Tony Petzold
14	2025/08/04	New function is added: LIS over TCP/IP implemented with software version 1.5	Tony Petzold

Disclaimer:

While HUMAN provides detailed documentation describing the Laboratory Information System (LIS) interface of HumaStar 100/200/300SR, it is the sole responsibility of the laboratory to ensure the correct configuration of their LIS in accordance with the provided protocol. HUMAN does not assume any responsibility for the setup, modification, or validation of the LIS system used by the laboratory.

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1 General

With software version 1.5 the HumaStar 100/200/300SR software can work with an LIS in two different ways, either based on file exchange (like with previous software versions), see chapter *2 LIS based on File Exchange between host and analyser* or based on Host query with ASTM over TCP/IP (implemented with software version 1.5), see chapter *3 ASTM over TCP.*

2 LIS based on File Exchange between host and analyser

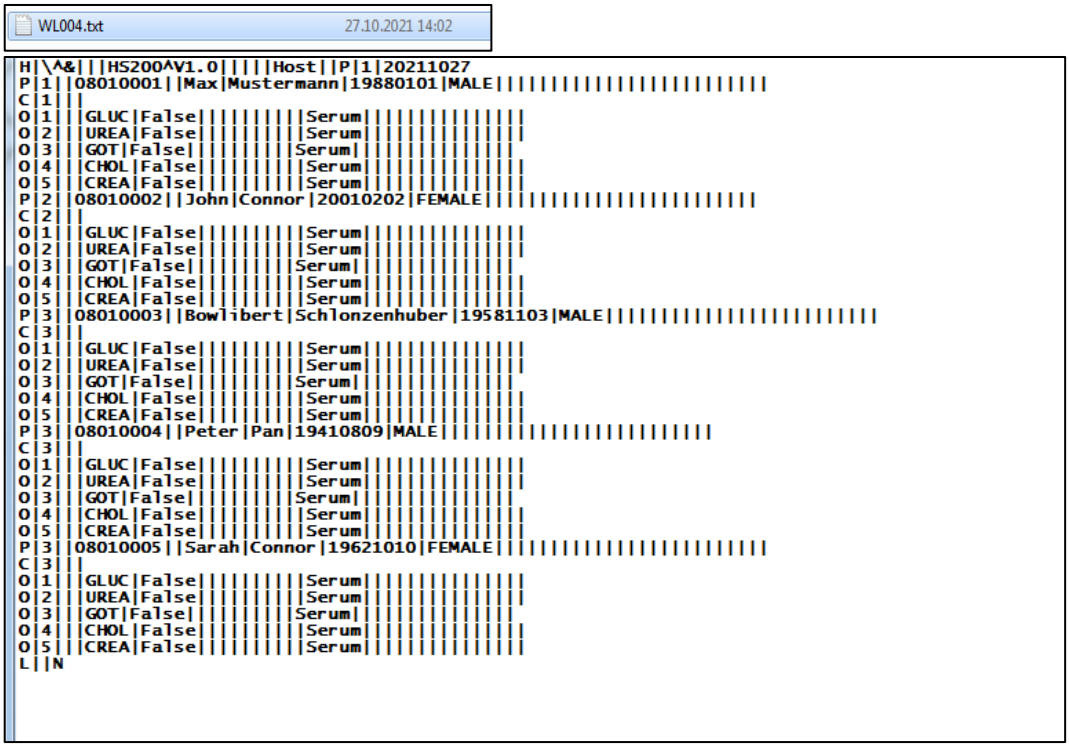
- the text files are encoded based on the ASTM standard
- the ASTM files are typically exchanged through the LAN, but the ASTM files can also be placed manually in the corresponding folder, see chapter *4 LIS ASTM work list folders*

2.1 Sequence of operation

1. Optional: Connect the PC of the analyzer to the host computer via LAN cable.

2. Create the worklist file in ASTM format (ASCII text file) with the extension *.astm* or *.txt*

Example:

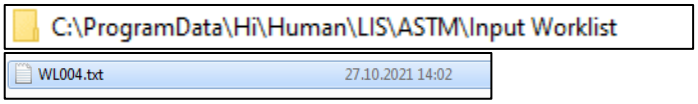


3. Transfer the worklist file to the folder **Input Worklist**

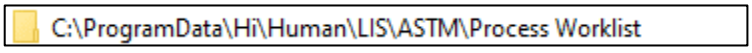
a) automatically from host computer via LAN

or

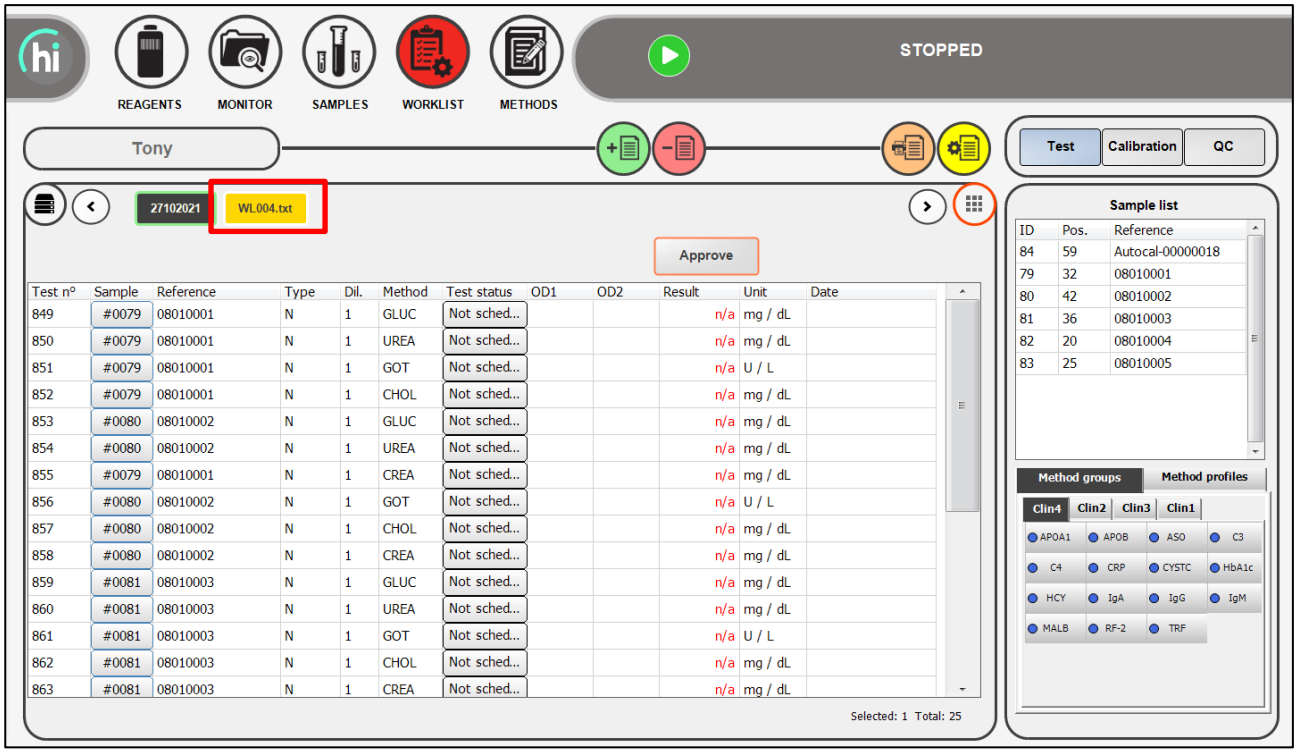
b) manually by copy & paste



4. Afterwards the analyzer recognizes the worklist file in the folder **Input Worklist** and moves the worklist file automatically to the folder **Process Worklist**



At the same time a new worklist is created in the software with the name of the worklist file.

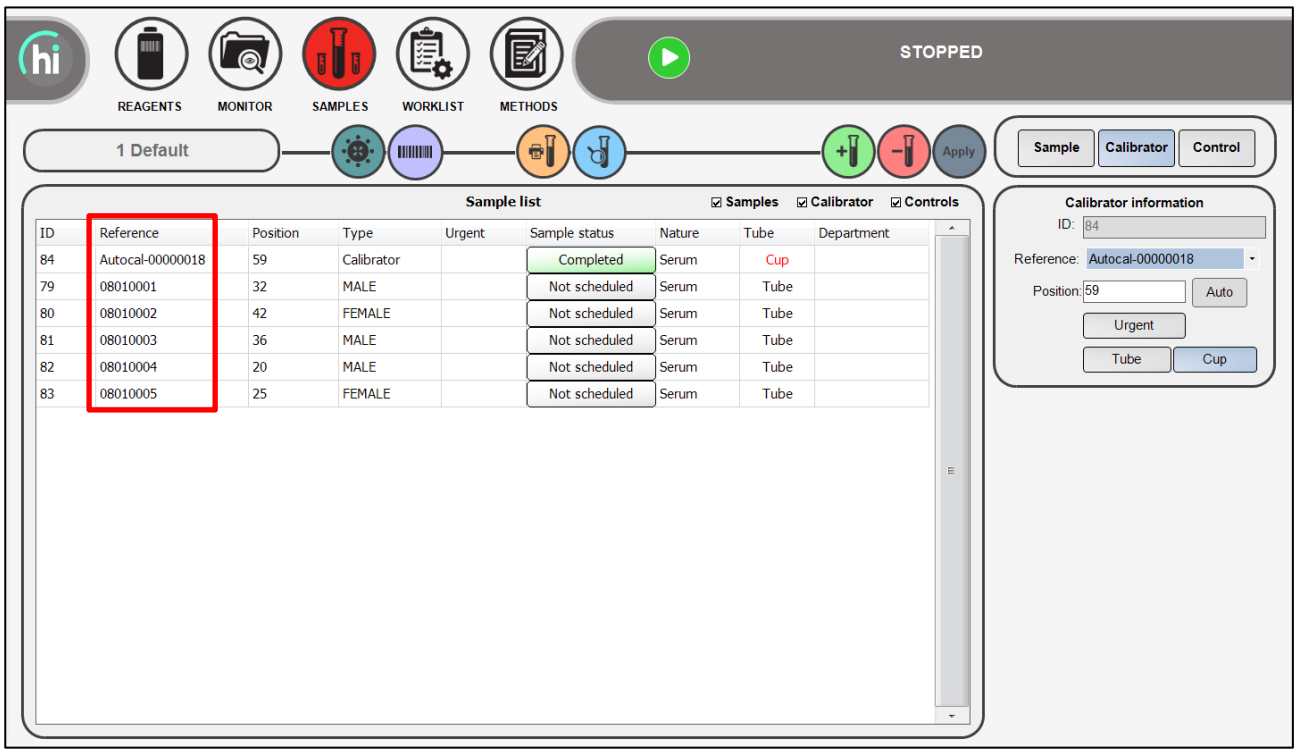


5. Navigate to **Samples**  in the software.

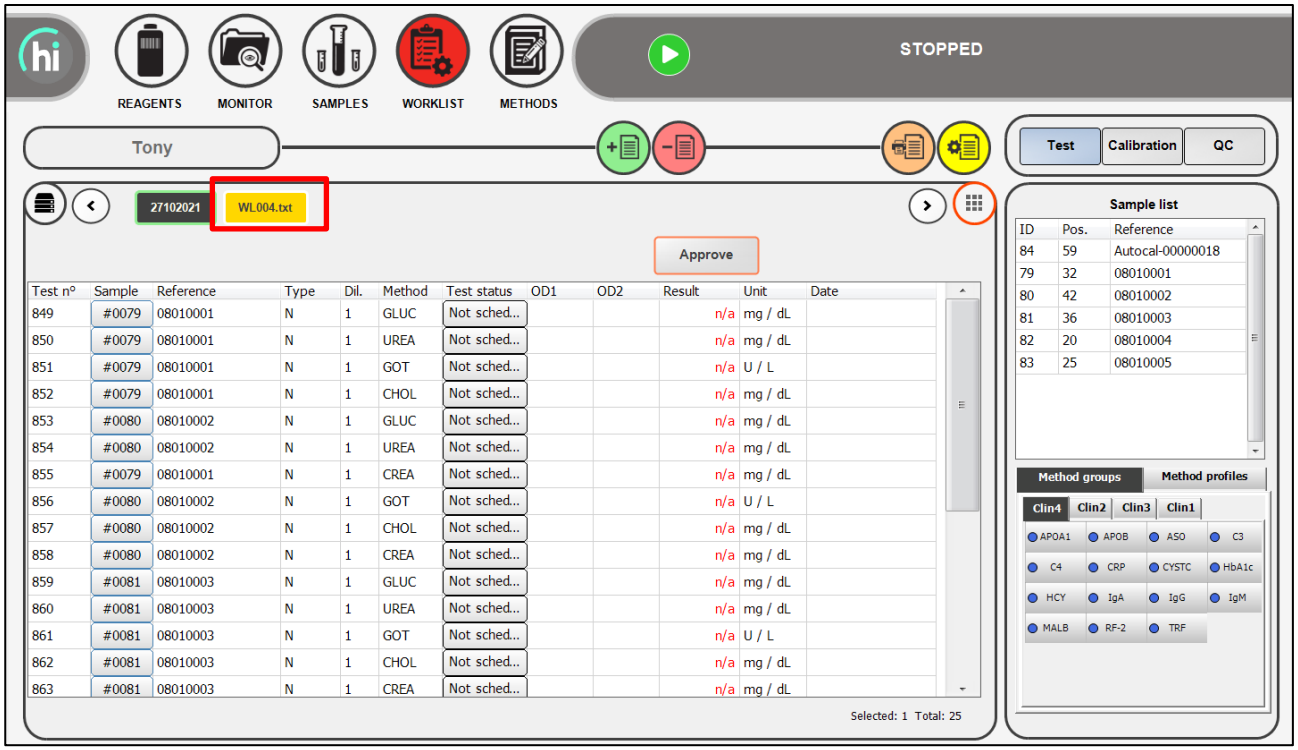
Then

a) manually assign the positions on the sample tray to the newly created samples

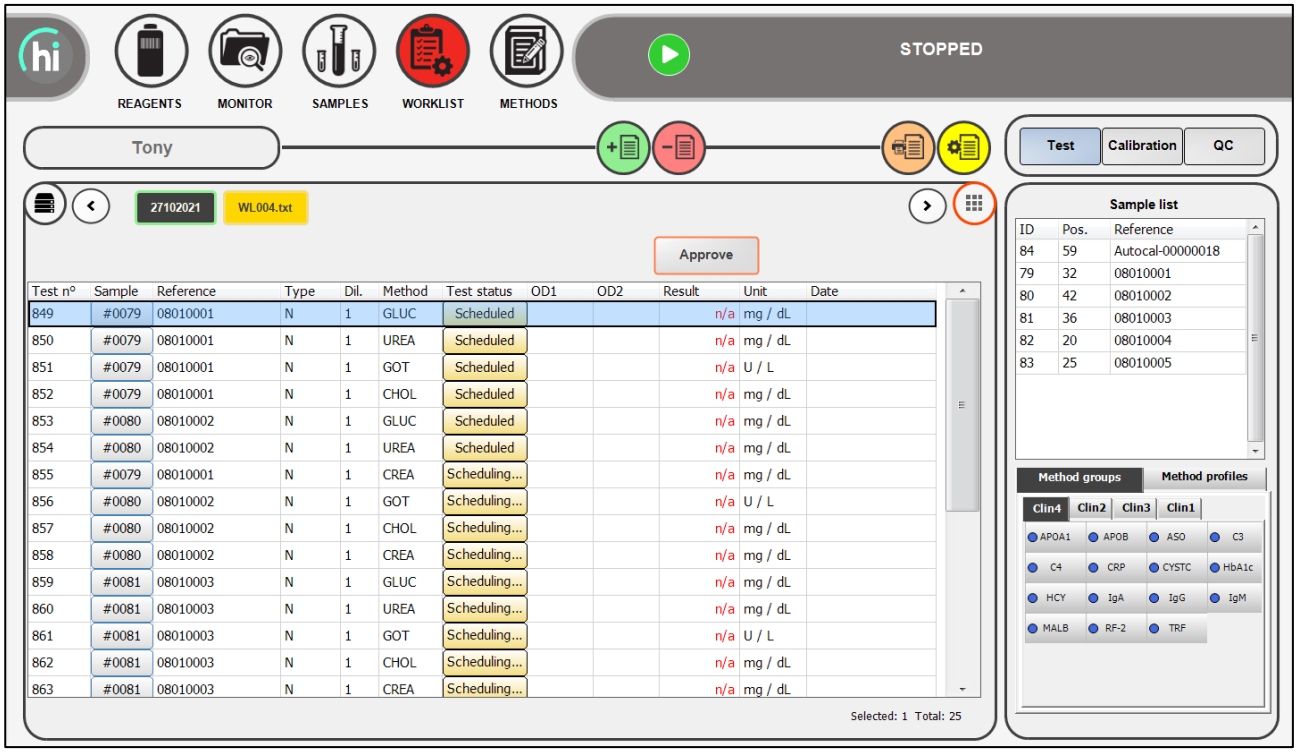
b) or in case barcode labelled samples are used, use the **Scan symbol**  to automatically assign the positions to the newly created samples. **Note:** The barcode number must be listed in **Field 4** of the **ASTM Input file.** Then the barcode number is transferred to the field “Reference” in the software.



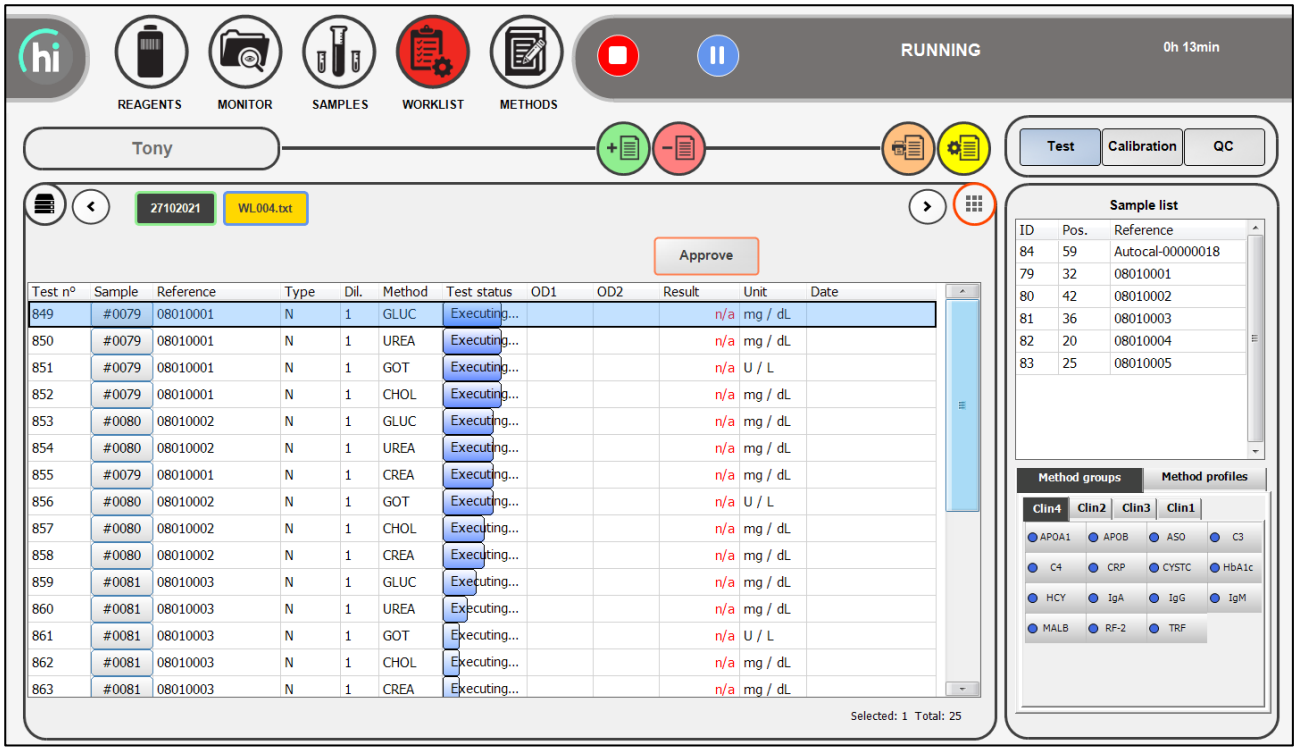
6. Navigate to **Worklist**  select the newly created **worksheet.**



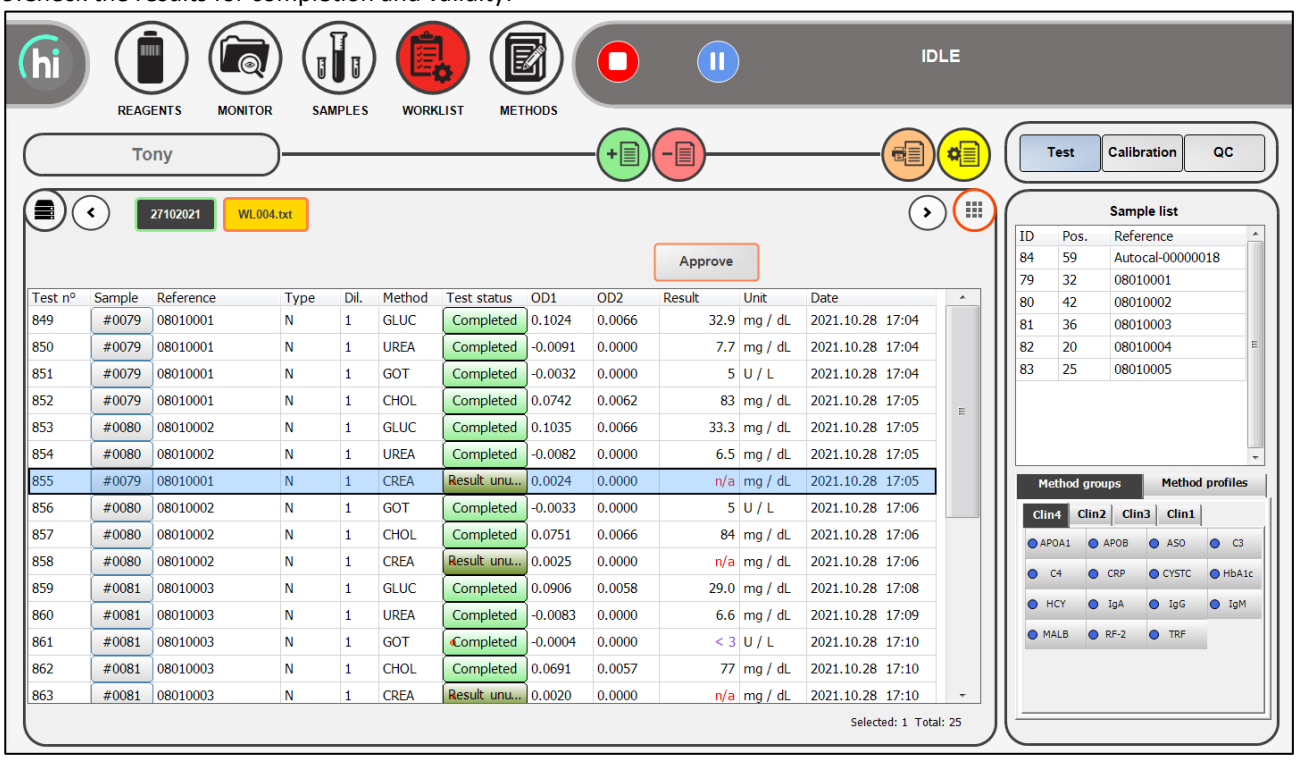
7. Check that all tests defined in the **worksheet** can be executed (reagents and samples on board). Then press the **Schedule button**.



8. The analyzer runs the tests.



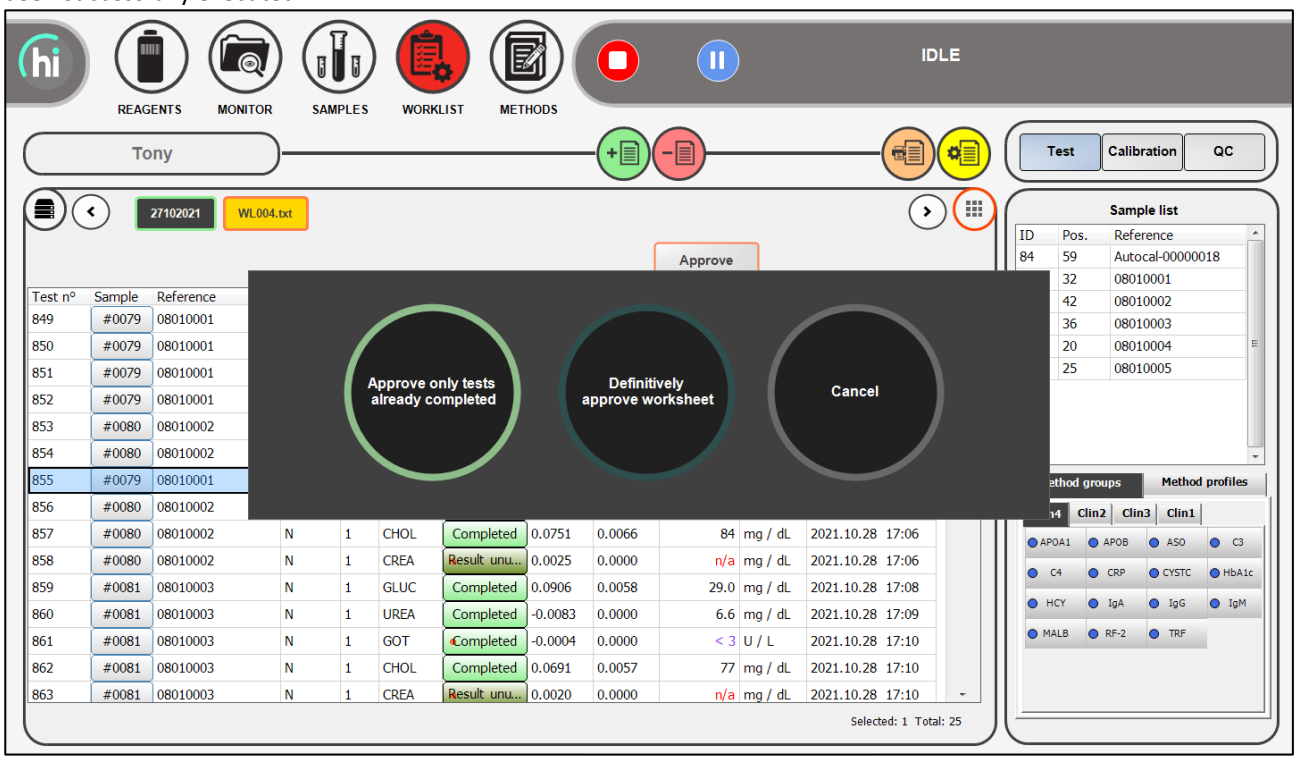
9.Check the results for completion and validity.



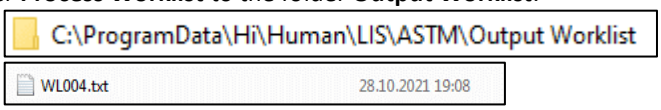
Finished results are automatically added to the **ASTM Process File** in the folder **Process Worklist.**

C:\ProgramData\Hi\Human\LIS\ASTM\Process Worklist

10. Press the **Approve button** to validate the worksheet. The worksheet can be validated even if not all the tests have been successfully executed.



11. The worksheet becomes **Approved** ✅ in the software and the **ASTM Process File** is automatically transferred from the folder **Process Worklist** to the folder **Output Worklist**.



12. The **ASTM Result File** containing the results can be transferred automatically to the host computer via LAN or it can be transferred manually to the corresponding folder of the LIS.
See example of the **ASTM Result File** below.

```
H|\^&|||Sphera^V1.0|||||Host||P|1|20211028190824
 P|1||08010001||08010001|||MALE|||||||||||||||||||||||||
  O|1|||GLUC|False|||||||||Serum|||||||||||||||
   R|1|GLUC||mg / dL||||32.9||||20211028170422|
  O|2|||UREA|False|||||||||Serum|||||||||||||||
   R|1|UREA||mg / dL||||7.7||||20211028170431|
  O|3|||GOT|False|||||||||Serum|||||||||||||||
   R|1|GOT||U / L||||5||||20211028170447|
  O|4|||CHOL|False|||||||||Serum|||||||||||||||
   R|1|CHOL||mg / dL||||83||||20211028170505|
 P|2||08010002||08010002|||FEMALE|||||||||||||||||||||||||
  O|1|||GLUC|False|||||||||Serum|||||||||||||||
   R|1|GLUC||mg / dL||||33.3||||20211028170533|
  O|2|||UREA|False|||||||||Serum|||||||||||||||
   R|1|UREA||mg / dL||||6.5||||20211028170541|
  O|3|||GOT|False|||||||||Serum|||||||||||||||
   R|1|GOT||U / L||||5||||20211028170626|
  O|4|||CHOL|False|||||||||Serum|||||||||||||||
   R|1|CHOL||mg / dL||||84||||20211028170634|
 P|3||08010003||08010003|||MALE|||||||||||||||||||||||||
  O|1|||GLUC|False|||||||||Serum|||||||||||||||
   R|1|GLUC||mg / dL||||29.0||||20211028170803|
  O|2|||UREA|False|||||||||Serum|||||||||||||||
   R|1|UREA||mg / dL||||6.6||||20211028170934|
  O|3|||GOT|False|||||||||Serum|||||||||||||||
```

2.3 Features

- Every worklist file placed in the folder **Input Worklist** will create a separate worksheet in the software.
- Several worksheets in the software can be executed at the same time.
- Several host computers can load worklist files at the same time into the software and for each worklist file a separate worksheet is created in the software.

2.4 LIS ASTM work list folders

The LIS ASTM work list folders are placed in the following folder:

- **Windows XP:**
  \Documents and Settings\All Users\Application data\HI\Human\LIS\ASTM.
- **Windows Vista / Windows Seven / Windows 10:**
  \ProgramData\HI\Human\LIS\ASTM.

Inside this \LIS\ASTM folder, there are three sub-folders:

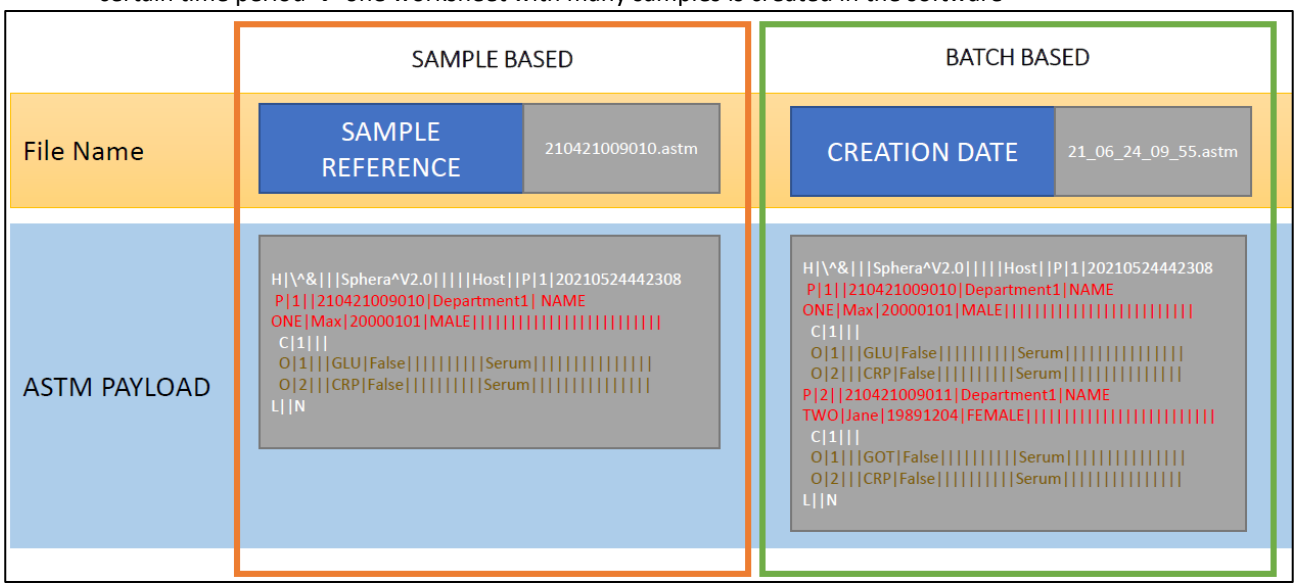
- The Input Folder ("\Input Worklist")
- The Process Folder ("\Process Worklist")
- The Output Folder ("\Output Worklist") - will be created when the results are approved for the first time

Note: No other folders must be generated in this folder structure! If additional folders are necessary for the development of the driver program, they have to be outside the HI folder and its sub-folders.

2.5 ASTM file types created by the LIS

It is possible to structure the ASTM files in two different types (must be done by the LIS company):

- Sample based type: each sample is represented by one worklist file → one worksheet per sample is created in the software
- Batch based type: the worklist file can contain more than one sample, the worklist file is created based on a certain time period → one worksheet with many samples is created in the software



Note: The assay name in the worklist file must exactly match the name of the same assay in the software.
For example: "Xyz" in the worklist file does not match "XYZ" with the assay name in the software.
For each assay, whose name does not exist in the software, cannot be transferred from worklist file to software.
Note: It is essential, that all the input worklist files have different names. Otherwise, they can be overwritten and some previous data can be lost.

2.6 ASTM standard input files encoding

H REPEAT_DELIMITER ('|') COMPONENT_DELIMITER ESCAPE_DELIMITER ||||||||||||

P | Sequence Number || Barcode | Department (optional) | Name (optional) | Family Name (optional) | Birthday | Type (optional) ||||||||||||||||||||||||

C | Sequence Number || Comments (all line optional) |

O | Sequence Number ||| Method Name | Is Urgent (true/false) |||||||||| Nature (Serum/Urine) ||||||||||||||

L ||N

Repeat delimiter ID defined on the first field of the header (second character) (default '|').
The field Birthday in the P record is mandatory. If this field is empty, the Birthday in the HI software becomes the day of the import, e.g. 20230830, after importing the worklist file to the HI software.

For a QC request (HumaStar 300SR only) the P record is following:
P | Sequence Number || Control name || Control name ||| CONTROL ||||||||||||||||||||||||

The following characters are used in the ASTM file but only for creating the ASTM file structure. Hence, they should not be used in the information of the test or sample.

|	U+007C	(field delimiter)
\	U+005C	(repeat delimiter)
^	U+005E	(component delimiter)
&	U+0026	(escape delimiter)
CR	U+000D	(line delimiter)
LF	U+000A	(line delimiter)

2.7 ASTM standard output files encoding

H REPEAT_DELIMITER ('|') COMPONENT_DELIMITER ESCAPE_DELIMITER ||| Software^version ||||| Host || P | 1 | DATE

P | Sequence Number || Barcode | Department (optional) | Name (optional) | Family Name (optional) | Birthday (optional) | Type (optional) ||||||||||||||||||||||||

C | Sequence Number || Sample.Comment |

O | Sequence Number ||| Method Name | Priority (true/false) |||||||||| Nature (Serum/Urine) ||||||||||||||

R | Sequence Number | Method Name |||||| Test Result |||| Test Completion Date |

L || N

The following characters are used in the ASTM file but only for creating the ASTM file structure. Hence they should not be used in the information of the test or sample.

|	U+007C	(field delimiter)
\	U+005C	(repeat delimiter)
^	U+005E	(component delimiter)
&	U+0026	(escape delimiter)
CR	U+000D	(line delimiter)
LF	U+000A	(line delimiter)

2.8 Examples

2.8.1 Example of ASTM work list input file

```
H|\^&|||HS300^V1.0|||||Host||P|1|20211027
P|1||08010001||Max|Mustermann|19880101|MALE|||||||||||||||||||||||||
C|1|||
O|1|||GLUC|False|||||||||Serum|||||||||||||||
O|2|||UREA|False|||||||||Serum|||||||||||||||
O|3|||GOT|False|||||||||Serum|||||||||||||||
O|4|||CHOL|False|||||||||Serum|||||||||||||||
O|5|||CREA|False|||||||||Serum|||||||||||||||
P|2||08010002||John|Connor|20010202|MALE|||||||||||||||||||||||||
C|2|||
O|1|||GLUC|False|||||||||Serum|||||||||||||||
O|2|||UREA|False|||||||||Serum|||||||||||||||
O|3|||GOT|False|||||||||Serum|||||||||||||||
O|4|||CHOL|False|||||||||Serum|||||||||||||||
O|5|||CREA|False|||||||||Serum|||||||||||||||
P|3||08010003||Bowlibert|Schlonzenhuber|19581103|MALE|||||||||||||||||||||||||
C|3|||
O|1|||GLUC|False|||||||||Serum|||||||||||||||
O|2|||UREA|False|||||||||Serum|||||||||||||||
O|3|||GOT|False|||||||||Serum|||||||||||||||
O|4|||CHOL|False|||||||||Serum|||||||||||||||
O|5|||CREA|False|||||||||Serum|||||||||||||||
L||N
```

2.8.2 Example of ASTM results output file

This example fits to the above input file. The work list was simply scheduled and approved, i. e. no measurement took place.

```
H|\^&|||Sphera^V1.0|||||Host||P|1|20160920091032
  P|1||00004|Department1|Max|Mustermann|20000000|MALE|||||||||||||||||||||||||
    C|1|||
    O|1|||Alb|False|||||||||Serum|||||||||||||||
      R|1|Alb||g/dl||||-99000000000||||00010101000000|
    O|2|||Amy|False|||||||||Serum|||||||||||||||
      R|1|Amy||U/l||||-99000000000||||00010101000000|
    O|3|||Bilda|False|||||||||Serum|||||||||||||||
      R|1|Bilda||mg/dl||||-99000000000||||00010101000000|
    O|4|||Bilta|False|||||||||Serum|||||||||||||||
      R|1|Bilta||mg/dl||||-99000000000||||00010101000000|
    O|6|||Chol|False|||||||||Serum|||||||||||||||
      R|1|Chol||mg/dl||||-99000000000||||00010101000000|
  P|2||00005|Department2|John|Connor|20010000|FEMALE|||||||||||||||||||||||||
    C|2|||
    O|1|||CK|False|||||||||Serum|||||||||||||||
      R|1|CK||U/l||||-99000000000||||00010101000000|
    O|2|||CK-MB|False|||||||||Serum|||||||||||||||
      R|1|CK-MB||U/l||||-99000000000||||00010101000000|
    O|3|||Cl|False|||||||||Serum|||||||||||||||
      R|1|Cl||mmol/L||||-99000000000||||00010101000000|
    O|4|||CreaA|False|||||||||Serum|||||||||||||||
      R|1|CreaA||mg/dl||||-99000000000||||00010101000000|
    O|6|||Glu|False|||||||||Serum|||||||||||||||
      R|1|Glu||mg/dl||||-99000000000||||00010101000000|
  P|3||00006|Department3|Bowlibert| Schlonzenhuber|20020000|MALE|||||||||||||||||||||||||
    C|3|||
    O|1|||LDH|False|||||||||Serum|||||||||||||||
      R|1|LDH||U/l||||-99000000000||||00010101000000|
    O|2|||LDL|False|||||||||Serum|||||||||||||||
      R|1|LDL||mg/dl||||-99000000000||||00010101000000|
    O|4|||C3|False|||||||||Serum|||||||||||||||
      R|1|C3||mg/dl||||-99000000000||||00010101000000|
    O|5|||C4|False|||||||||Serum|||||||||||||||
      R|1|C4||mg/dl||||-99000000000||||00010101000000|
   O|6|||Hba1C|False|||||||||Serum|||||||||||||||
      R|1|Hba1C||mmol/mol Hb||||-99000000000||||00010101000000|
L||N
```

2.8.3 Example for usage of special characters

In this example there is a special character "ü" used in the Family Name in the work list input file.

```
P|2||00005|Department2|Lieschen|Müller||20010202|FEMALE|||||||||||||||||||||||||
```

The corresponding results output file cannot correctly operate this special character.

```
P|2||00005|Department2|Lieschen|Mi¿½ller|20010000|FEMALE|||||||||||||||||||||||||
```

To avoid this situation, do not use special characters. The following workaround is possible in this example.

```
P|2||00005|Department2|Lieschen|Mueller|20010202|FEMALE|||||||||||||||||||||||||
```

2.8.4 Example of ASTM work list input file with QC request (HumaStar 300SR only)

```
H|\^&|||HS300SR^V1.0|||||Host||P|1|20190607134056
 P|1||0000000001||0000000001|||MALE|||||||||||||||||||||||||
  O|1|||GOT|False|||||||||Serum|||||||||||||||
 P|2||Serodos-00000004||Serodos-00000004|||Control|||||||||||||||||||||||||
  O|1|||GOT|False|||||||||Serum|||||||||||||||
L||N
```

2.8.5 Example of ASTM work list input file with QC request (HumaStar 300SR only)

This example fits to the above input file. The work list was simply manually edited and approved, i. e. no measurement took place.

```
H|\^&|||HS300SR^V1.0|||||Host||P|1|20190607134056
 P|1||0000000001||0000000001|||MALE|||||||||||||||||||||||||
  O|1|||GOT|False|||||||||Serum|||||||||||||||
   R|1|GOT||U / l||||125.0||||20190607134012|
 P|2||Serodos-00000004||Serodos-00000004|||Control|||||||||||||||||||||||||
  O|1|||GOT|False|||||||||Serum|||||||||||||||
   R|1|GOT||U / l||||12.0||||20190607134042|
L||N
```

3 ASTM over TCP

3.1 Standards

The HI Software implements a **subset** of the ASTM protocol according to the following standards:

- LIS1-A Formerly ASTM E1381-02, *Standard Specification for Low-Level Protocol to Transfer Messages Between Clinical Laboratory Instruments and Computer Systems*
- LIS2-A2 Replaces LIS2-A, *Specification for Transferring Information Between Clinical Laboratory Instruments and Information Systems; Approved Standard—Second Edition*

Any future implementation will follow these standards.

3.2 TCP

When enabled, the HI Software ASTM service will listen on port 6557 (default value).
Note: Firewalls must be configured accordingly to enable communication on the ASTM port. The HI Software may use Secure Socket Layer (SSL with a self-signed certificate) to encrypt data on the TCP port.

3.3 Transport Layer

ASTM must be transmitted with ASCII encoding.
ASTM payload should not use any of the listed control characters except <CR>.

Table 1: ASCII control characters

In use		Reserved	
STX	2	SOH	1
ETX	3	DLE	16
EOT	4	DC1	17
ENQ	5	DC2	18
ACK	6	DC3	19
LF	10	DC4	20
CR	13	SYN	22
NAK	21	ETB	23
		ESC	27

3.3.1 Establishment phase

Any client initiates the conversation transmitting <ENQ>.
Server will reply with:

- <ACK> - if the server is ready initiating he Transfer Phase
- <NAK> - if the server is busy, in that case the client must wait at least 10 seconds before retransmitting <ENQ>

3.3.2 Transfer phase

The HI Software does not use ASTM multi-Frames, all the ASTM data must be included in a single frame.

Frame structure:
<**STX**> [FN] ASTM <**ETX**> C1 C2 <**CR**> <**LF**>

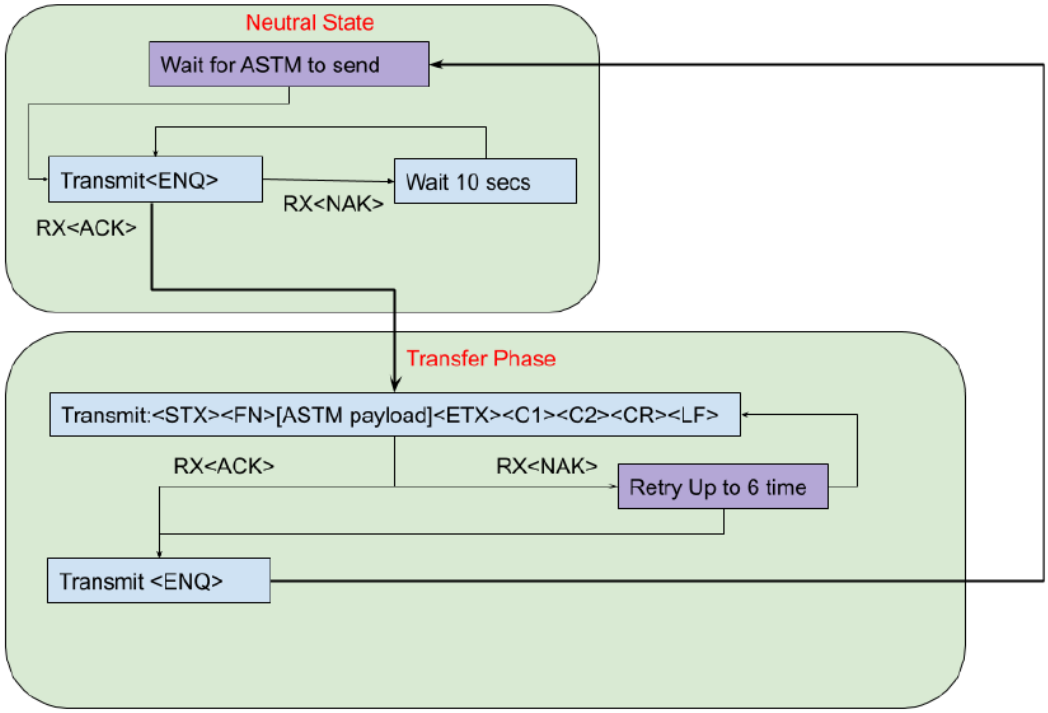
where:
<**STX**> = Start of Text transmission control character
FN = single digit Frame Number 0 to 7 (always 0)
ASTM = ASTM Data Content of Message
<**ETX**> = End of Text transmission control character
C1 = most significant character of checksum 0 to 9 and A to F
C2 = least significant character of checksum 0 to 9 and A to F
<**CR**> = Carriage Return ASCII character
<**LF**> = Line Feed ASCII character

Frame Number:
The frame number permits the receiver to distinguish between new and retransmitted frames. It is a single digit sent immediately after the <STX> character, always '0' (ASCII 48) for HI Software.

Checksum:
The checksum is encoded as two characters which are sent after the <ETX> character. The checksum is computed by adding the binary values of the characters, keeping the least significant eight bits of the result. The checksum is initialized to zero with the <STX> character.

The first character used in computing the checksum is the frame number. Each character in the message text is added to the checksum (modulo 256). The computation for the checksum does not include <STX>, the checksum characters, or the trailing <CR> and <LF>.The checksum is an integer represented by eight bits, it can be considered as two groups of four bits. The groups of four bits are converted to the ASCII characters of the hexadecimal representation. The two ASCII characters are transmitted as the checksum, with the most significant character first.

The server replies to the request with:
- <ACK> - signifies the last frame was received successfully
- <NAK> - signifies the last frame was not successfully received



3.3.3 Termination phase

The sender transmits the <EOT> transmission control character to communicate the end of transmission to the server, Data Link is back to Neutral State.

3.4 ASTM Payload (High Level Control)

In general, the ASTM payload must be set according to the **LIS1-A** standard.
In specific, the following command record types are allowed.

Table 2: Allowed record types

Type	Level	Function Name
H	0	Message Header Record
P	1	Patient Information Record
C	2	Comment Record
O	2	Test Order Record
R	3	Results Record
L	0	Message Terminator Record

Note: The sequence number applies to all command records and it shall be generated for every message. Initial value shall be 1. Every transmission increments the number by 1. The number will set back to 1 at transmission of separate command with same level or command with higher level.

3.4.1 H Message Header Record

	Designation	**Process upon reception from Host**	**Process on transmission from Analyzer**
1	Record type	Record type H	
2	Delimiters	| or \ or ^ or &	
3	MessageControlID	Optional LIS ID or sheet name	LIS ID
4	Access password	-	-
5	Sender ID	-	Instrument serial number
6	Sender Street Address	-	-
7	Reserved Field	-	-
8	Sender Telephone Number	-	-
9	Characteristics Sender	-	-
10	Receiver ID	-	-
11	Comment Special Instructions	-	-
12	Processing ID	-	P
13	Version Number	-	Hi^V2.1
14	Date Time Message	-	Date and time at transmission. YYYYMMDDhhmmss

3.4.2 P Patient Information Record

The string is sent from host to analyzer at ordering test requests. This is also sent from analyzer to host when sending test results which as specified by host.

	Designation	**Process upon reception from Host**	**Process on transmission from Analyzer**
1	Record type	Record type P	
2	Sequence Number	See note above.	
3	Practice ID	Patient Unique Identifier (optional), if empty no patient is imported	Send as received
4	Laboratory ID	Default Unique Sample ID, for	Send as received

		the order records (used when "Order SampleID" is not defined)	
5	Patient ID	Operational Department	Send as received
6	Patient Full Name	LastName^FirstName^Middle NameOrTitle	Send as received
7	Mother Name	Alternative LastName	Send as received
8	Birthday	Format is "YYYYMMDD"	Send as received
9	Sex	Used for Sample Type	Send as received
10	Address	-	-
11	Patient Address	-	-
12	Reserved	-	-
13	Telephone	-	-
14	Physician ID	-	-
15	Special Field 1	-	-
16	Special Field 2	-	-
17	Height	-	-
18	Weight	-	-
19	Suspected Diagnosis	-	-
20	Active Medications	-	-
21	Diet	-	-
22	Practice Field 1	-	-
23	Practice Field 2	-	-
24	Admission And Discharge	-	-
25	Admission Status	-	-
26	Location	-	-
27	Nature Alternative Diagnostic Code	-	-
28	Alternative Diagnostic Code	-	-
29	Religion	-	-
30	Marital Status	-	-
31	Isolation Status	-	-
32	Language	-	-
33	Hospital Service	-	-
34	Hospital Institution	-	-
35	Dosage Category	-	-

3.4.3 C Comment Record

	Designation	**Process upon reception from Host**	**Process on transmission from Analyzer**
1	Record type	Record type C	
2	Sequence Number	See note above	
3	Source of comment	-	-
4	Text of comment	Optional	Send as received
5	Type of comment	-	

3.4.4 O Test Order Record

	Designation	Process upon reception from Host	Process on transmission from Analyzer
1	Record type	Record type O	
2	Sequence Number	See note above	Send as received
3	Specimen ID	Specimen unique ID. The position of the sample can also be specified with the following structure ID^POSITION^Predilution (e.g. 00001^1^2) refers to the sample id 00001 to be placed in position 1 of the sample tray, with a predilution of 2)	Send and receive
4	Instrument Specimen ID	-	Send back Sample Id
5	Universal Test ID	Method unique code (not that the code should be identical to the Method NAME in HI software in order to be recognized)	Send as received
6	Priority	TRUE or “A” for urgent	Send “A” if urgent
7	Requested Ordered Date Time		
8	Specimen Collection Date Time		
9	Collection End Time		
10	Collection Volume		
11	Collector ID		
12	Action Code		
13	Danger Code		
14	Relevant Clinical Information		
15	Date Time Specimen Received		
16	Specimen Description	SERUM default or URINE	Send as received
17	Ordering Physician		
18	Physician Telephone Number		
19	User Field Number 1		
20	User Field Number 2		
21	Laboratory Field Number 1		
22	Laboratory Field Number 2		
23	Date Time Result Last Modified		
24	Instrument Charge Information System		
25	Instrument Section ID		
26	Report Types		
27	Reserved Field		
28	Location Specimen Collection		
29	Nosocomial Infection Flag		
30	Specimen Service		
31	Specimen Institution		

3.4.5 R Result Record

Test results of test items are notified. This record is process only by the Analyzer and should not be inserted into the ASTM payload by the HOST.

	Designation	Process upon reception from Host	Process on transmission from Analyzer
1	Record type	Record type R	
2	Sequence Number		See note above
3	Universal Test ID		Test Method Name
4	Data Measurement Value		Test result
5	Unit		Unit used for the measurement (the user can choose which unit is used in the software settings)
6	Reference Ranges		Pathology ranges, if defined. Example: “0.0 to 2.5”
7	Result Abnormal Flags		See table below
8	Nature Abnormality Testing		-
9	Result Status		-
10	Date Change Instrument Normative		-
11	Operator Identification		User ID
12	Date Time Test Started		-
13	Date Time Test Completed		YYYYMMDDhhmmss
14	Instrument Identification		Instrument Serial number

Table 3: Test result error flag code

Transmission Code	Description	Priority	Display
A	Error occurred	1	Error flag
>	Technical range over “High”	2	>
<	Technical range under “Low”	2	<
H	Normal range over “High”	3	H
L	Normal range under “Low”	3	L
N	Normal	4	N/A
LL	Unused	-	-
HH	Unused	-	-
U	Unused	-	-
D	Unused	-	-
B	Unused	-	-
W	Unused	-	-

3.4.6 Q Request Information Record

Not Supported.

3.4.7 S Scientific Record

Not Supported.

4 Additional notes